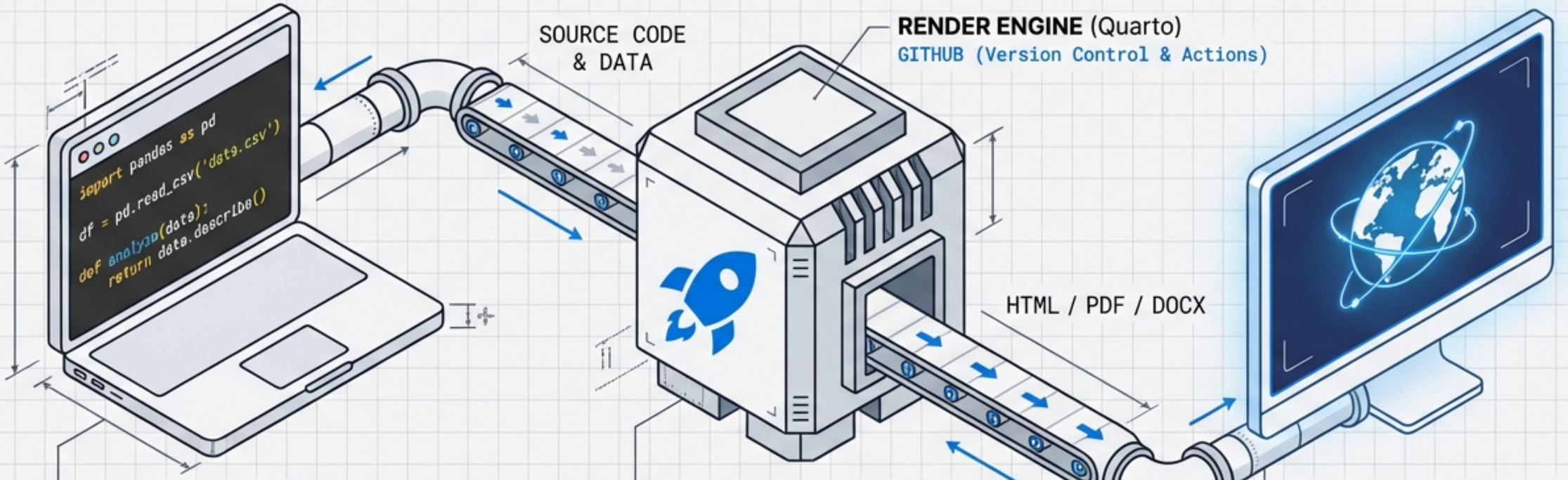
The Data Science Portfolio Stack

Understanding Why We Use Quarto, GitHub, and Virtual Environments



1 **CODE EDITOR** (Python)
Slate Grey JetBrains Mono
VIRTUAL ENVIRONMENTS

2 **RENDER ENGINE** (Quarto)
Slate Grey JetBrains Mono

3 **PUBLISHED WEBSITE** (Online)
Slate Grey JetBrains Mono
HOSTING & SHARING

Goal: Move from following instructions to understanding the architecture.

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You Are Building More Than Just a Script

Raw Script

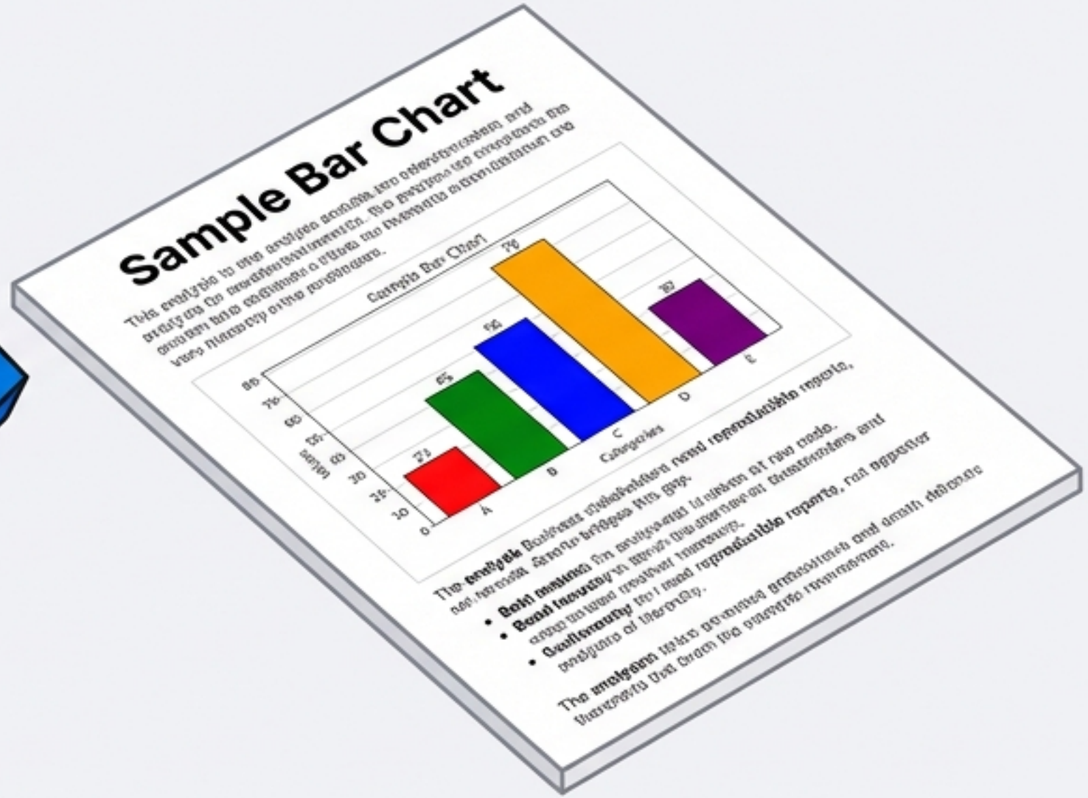
```python
import pandas as pd
import matplotlib.pyplot as plt

data = {'A': 23, 'B': 45, 'C': 56, 'D': 78, 'E': 32}
df = pd.DataFrame(data, index=[0])

plt.bar(data.keys(), data.values())
plt.title('Sample Bar Chart')
plt.show()
```

The Quarto Workflow

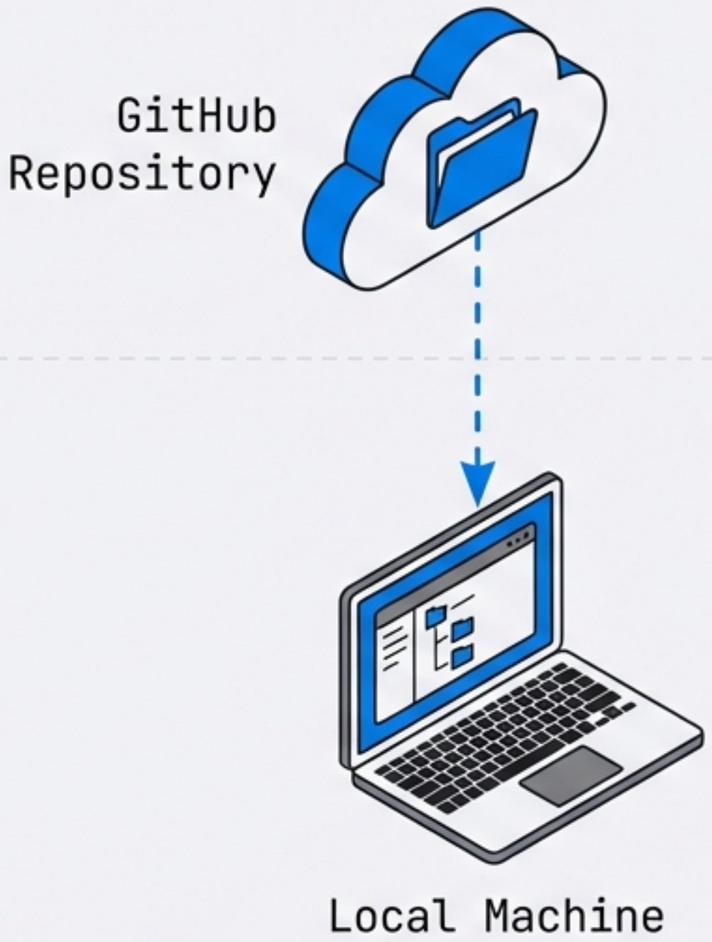
Professional Portfolio



The Why: Business stakeholders need reproductiable reports, not raw code. Quarto bridges this gap.

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The Foundation: GitHub & Repositories

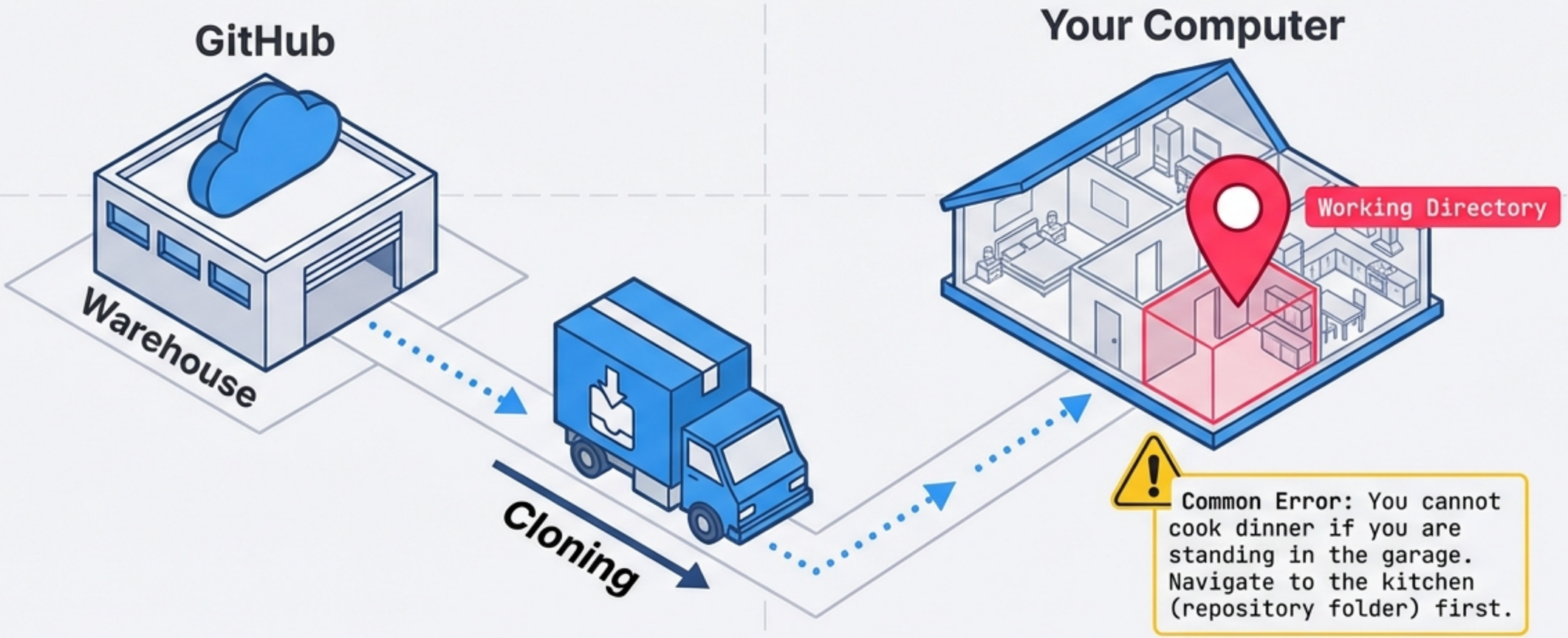


Why use a Repository?

- **Safety**: Cloud backup prevents data loss.
- **Portfolio**: Enables public hosting via `GitHub Pages`.
- **Organization**: Isolates project files from other coursework.

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Concept Check: Cloning & The Working Directory

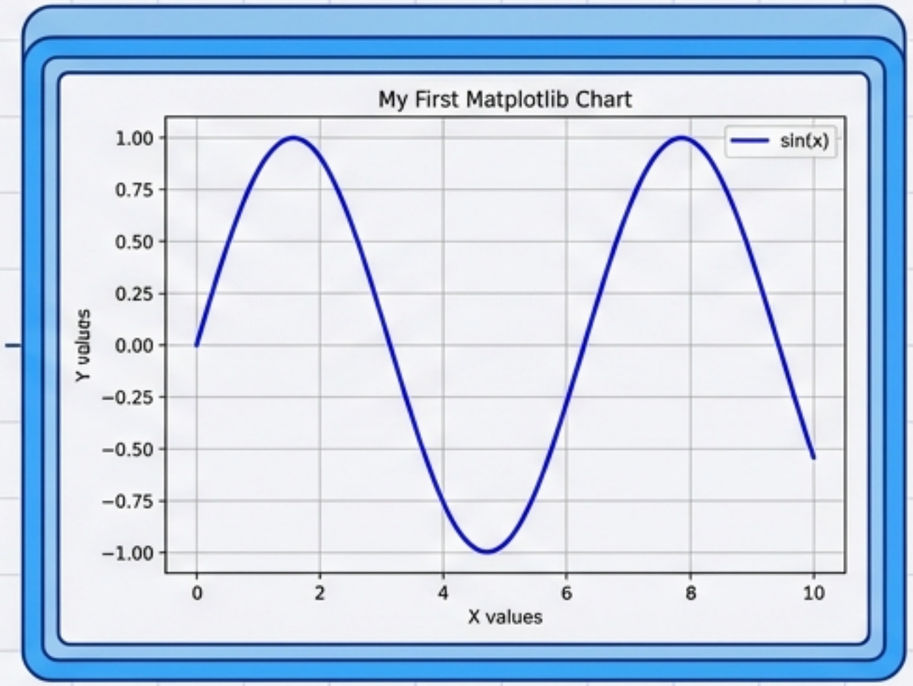


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The Engine: Python & Matplotlib

```
x = np.linspace(0, 10, 100)
y = np.sin(x)
plt.plot(x, y)
```

Automated Visualization

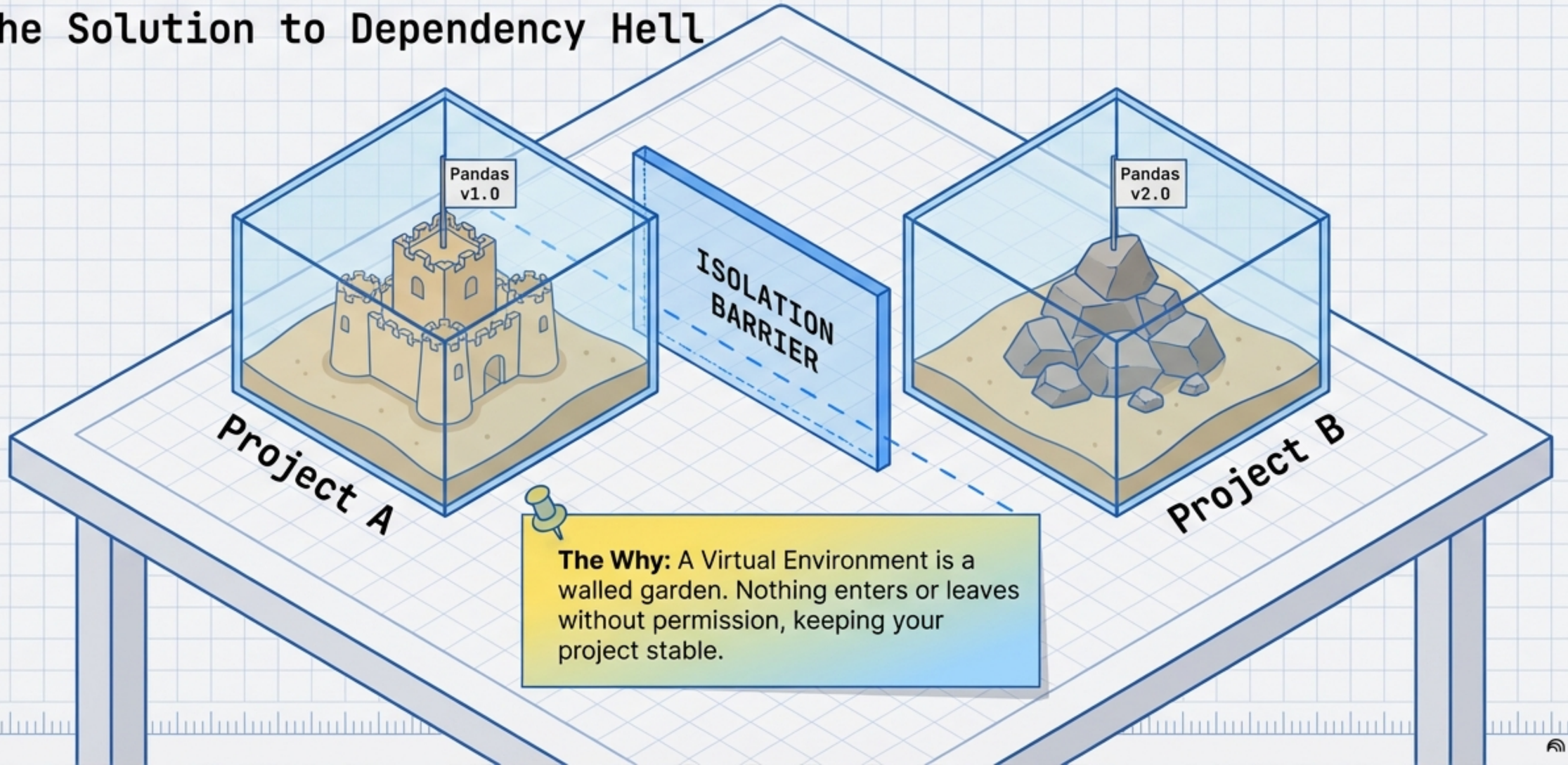


The Why: We script charts so they are reproducible. If data changes, the chart updates instantly.

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The Protection: Virtual Environments (venv)

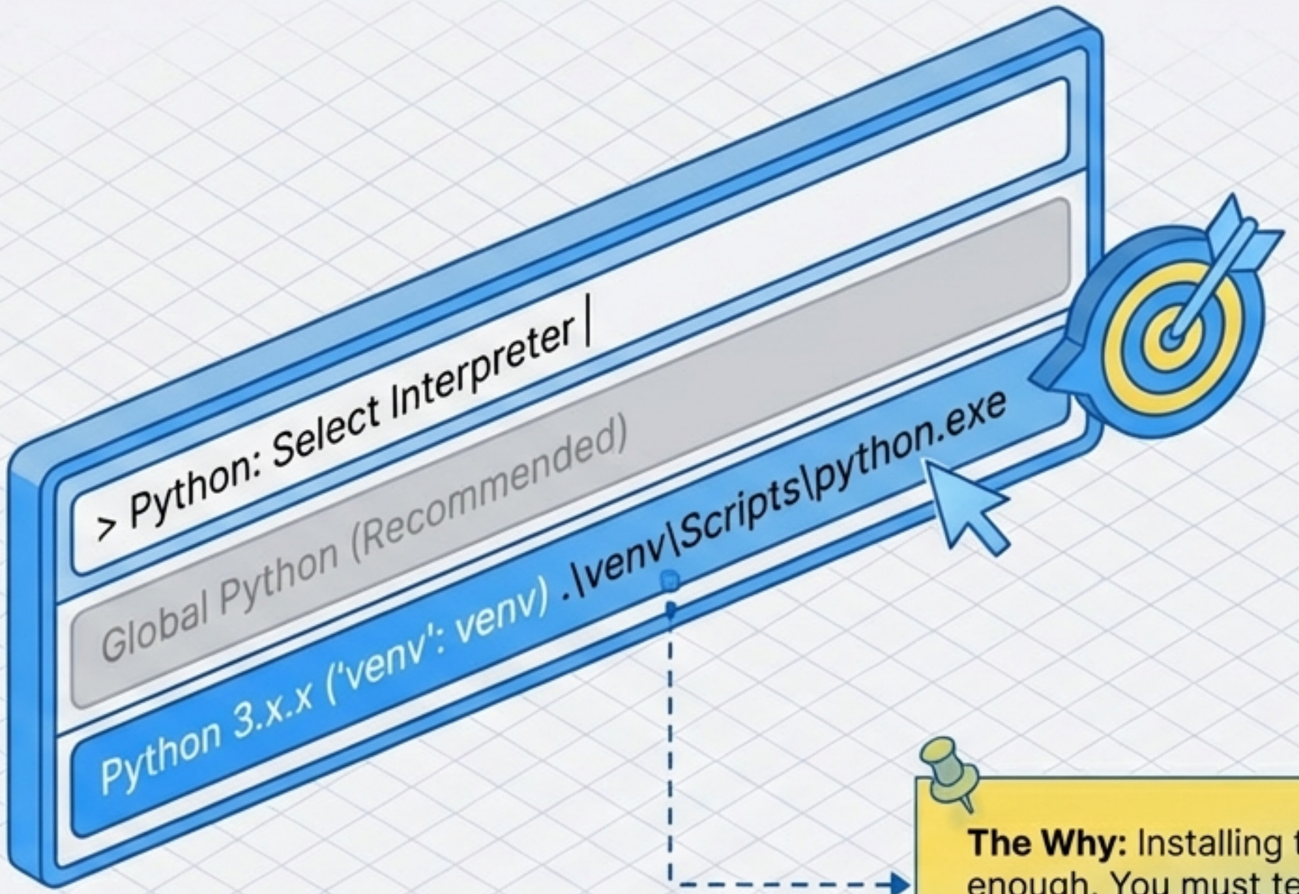
The Solution to Dependency Hell



The Why: A Virtual Environment is a walled garden. Nothing enters or leaves without permission, keeping your project stable.

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The Connection: Selecting the Interpreter



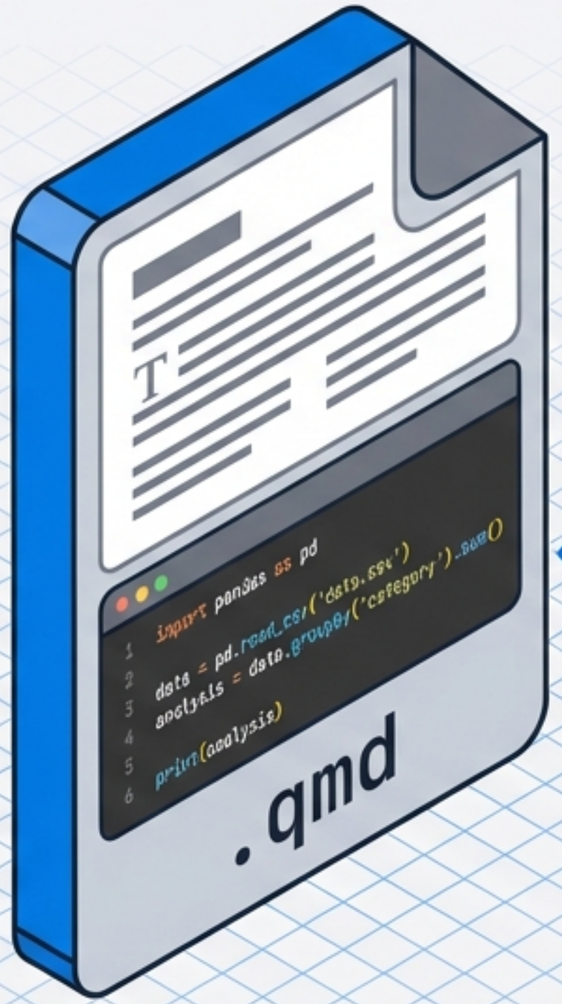
The Why: Installing the environment is not enough. You must tell your editor to *use* the specific tools inside the venv folder.

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Why Quarto?

The Integrator: What is Quarto?

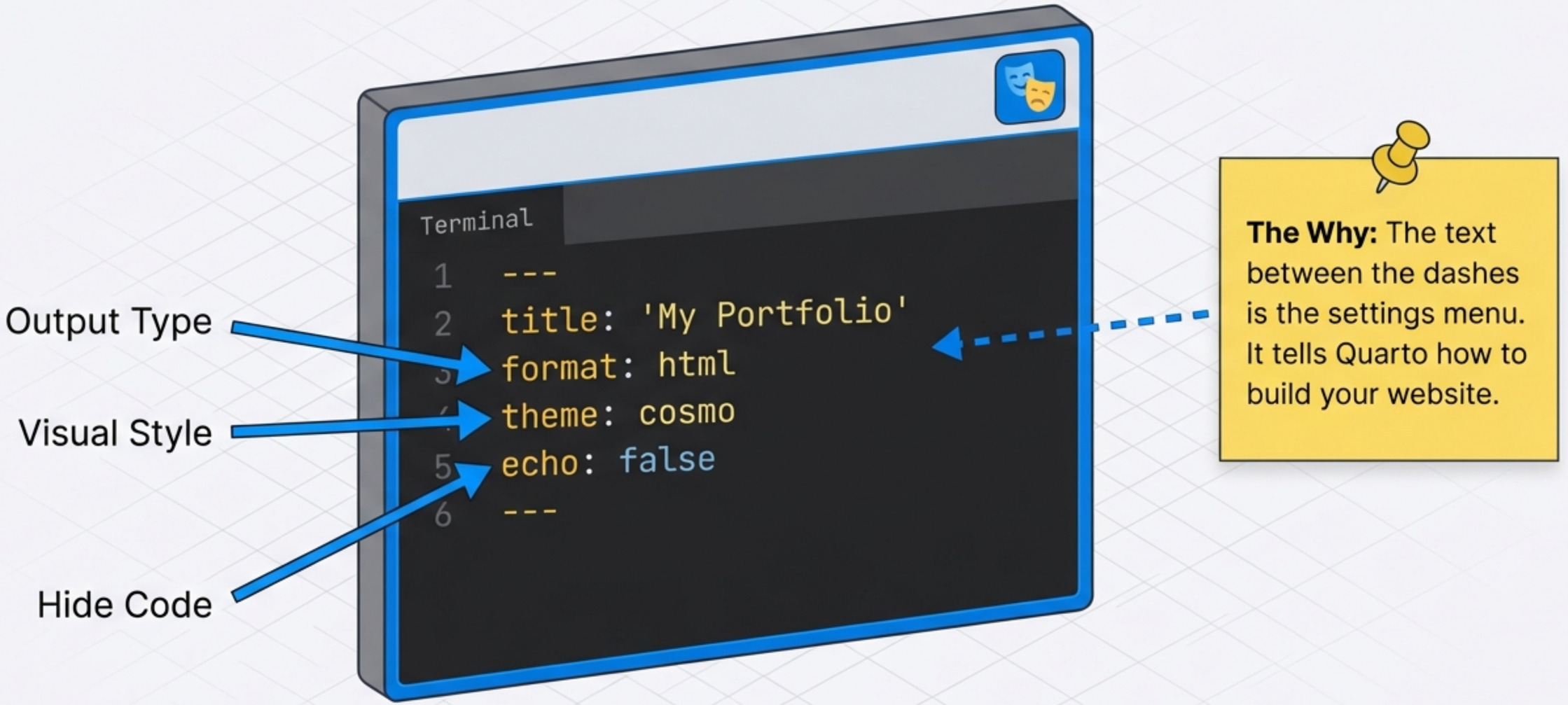
Literate Programming



The Why:
Quarto allows you to explain your logic (text) directly alongside your analysis (code) in a single file.

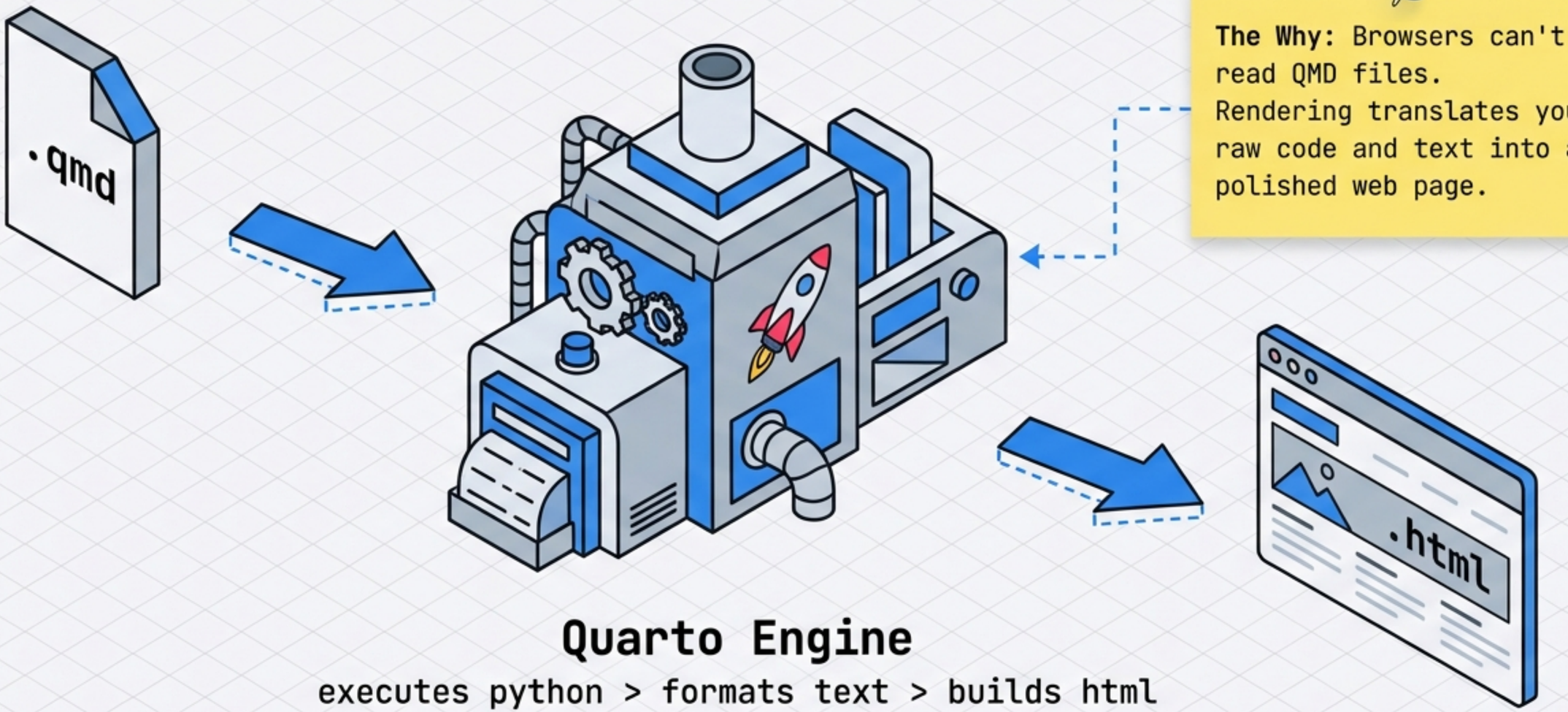
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The Control Panel: YAML Front-Matter



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The Transformation: Rendering



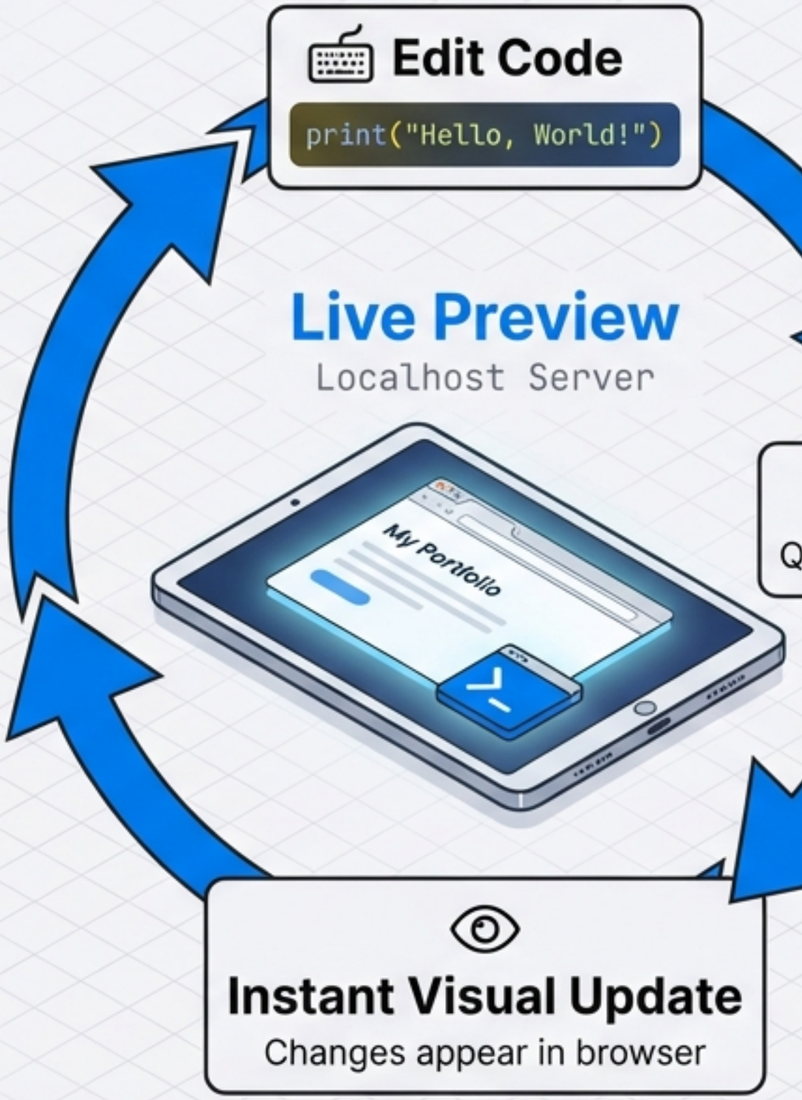
The Why: Browsers can't read QMD files. Rendering translates your raw code and text into a polished web page.

Quarto Engine

executes python > formats text > builds html

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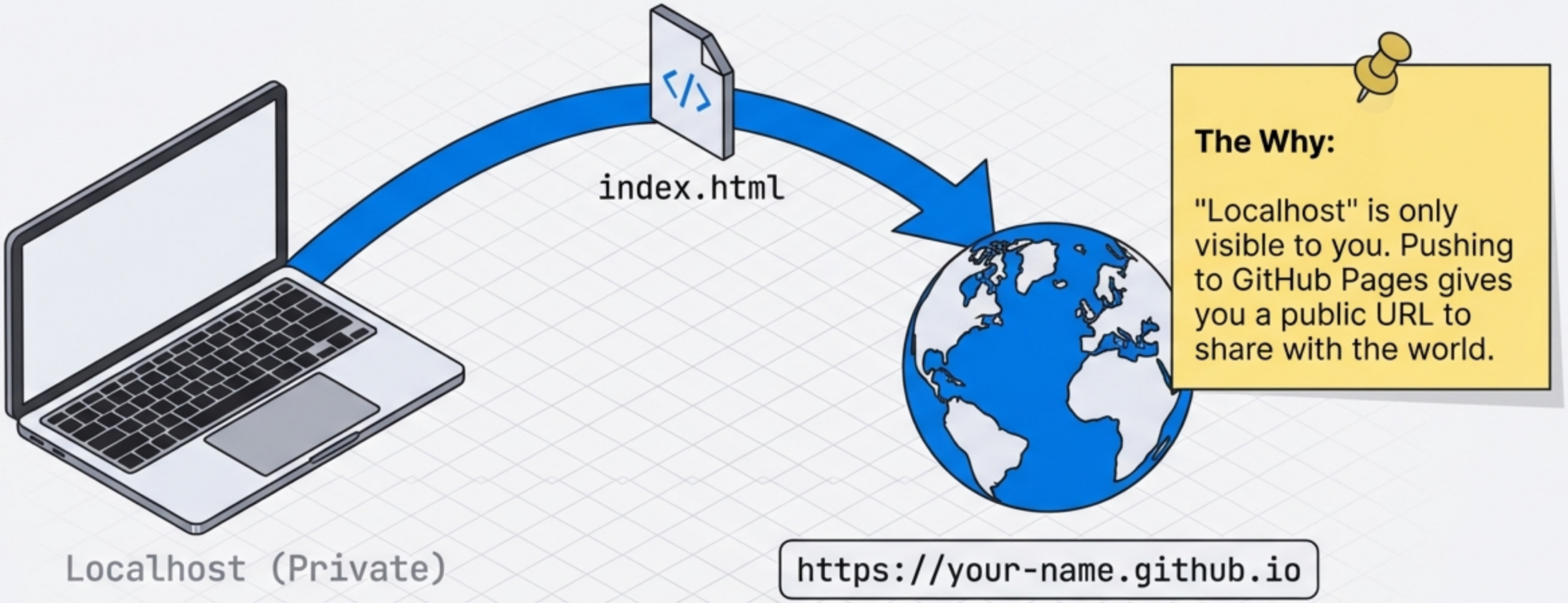
The Feedback Loop: Live Preview



The Why: Re-rendering the whole site takes time. Live Preview creates a temporary local server so you can see changes instantly.

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The Grand Finale: Publishing with GitHub Pages

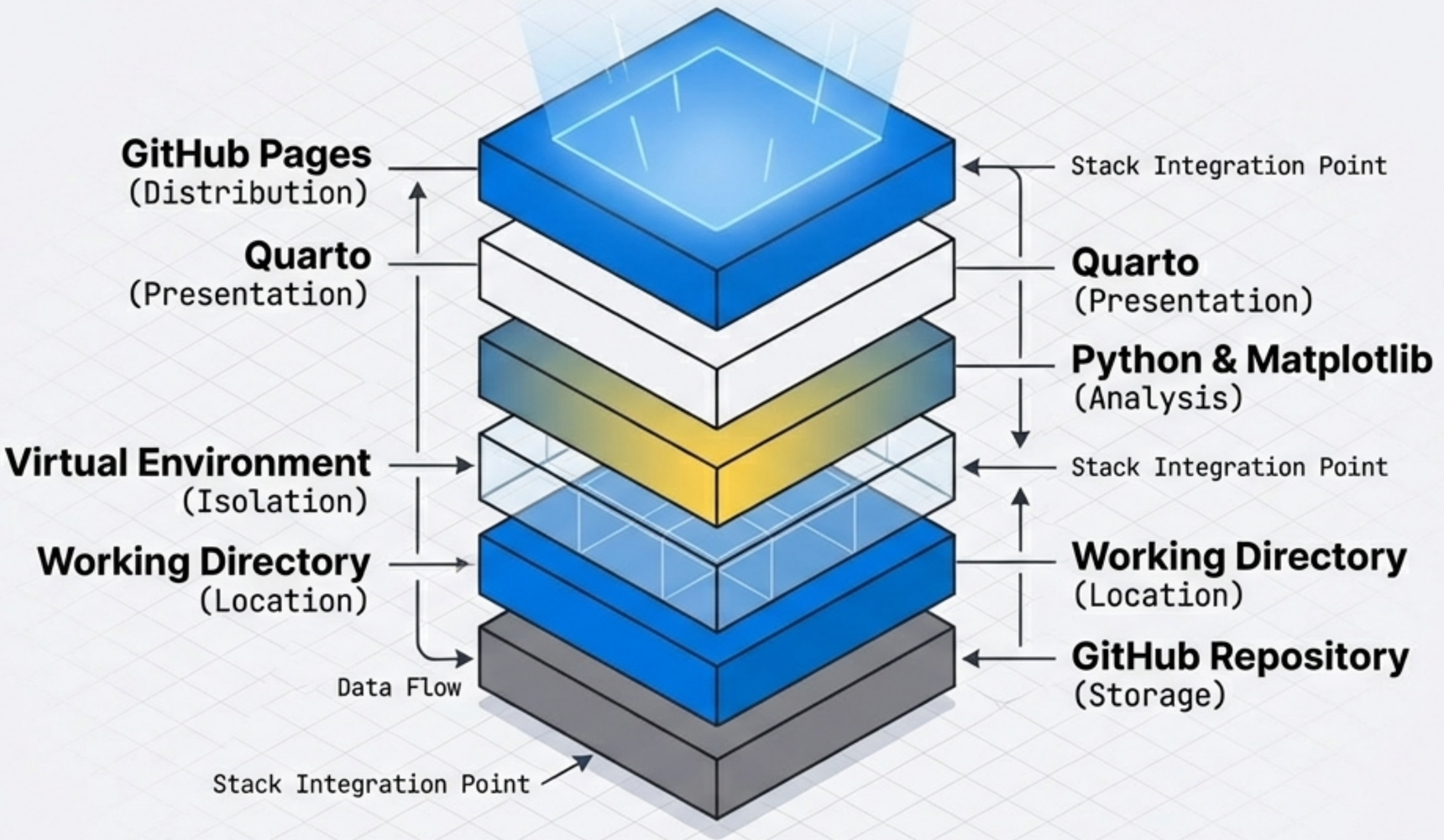


The Why:

"Localhost" is only visible to you. Pushing to GitHub Pages gives you a public URL to share with the world.

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Your Full Data Science Stack



Now that you know the *Why*, go build the *What*.

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